



CYPRUS INTERNATIONAL UNIVERSITY

Database Management Systems and ProgrammingII— Final Project 2026

Wildlife Conservation Management System

Using Supabase & PostgreSQL

Supabase

PostgreSQL

HTML/CSS/JS



Protecting Wildlife Through Intelligent Data Management

STUDENT CONTRIBUTIONS

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Why Wildlife Conservation Systems?



Wildlife populations worldwide face unprecedented threats from habitat loss, climate change, and poaching. Effective conservation requires robust data management to track animals, monitor sightings, and coordinate protection programs.



Objective

Build a full-stack system to manage animal data, sightings, conservation programs, and user roles in a structured PostgreSQL database.



Database Focus

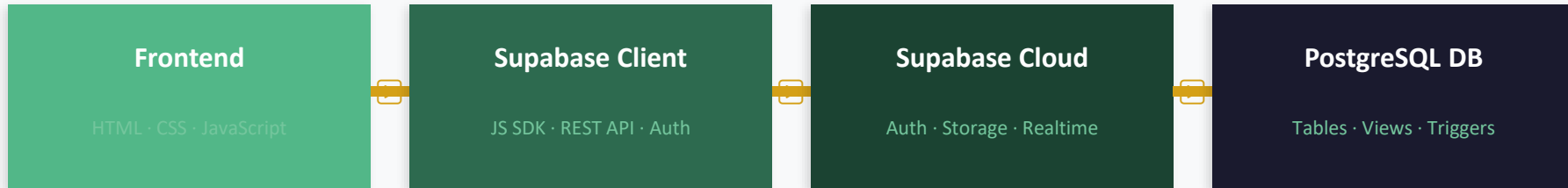
Leverage Supabase and PostgreSQL with advanced SQL features: triggers, views, stored procedures, and full-text search.



Assumptions

Users have internet access; roles are predefined (admin/ranger/researcher); all animal data is entered by authorized users only.

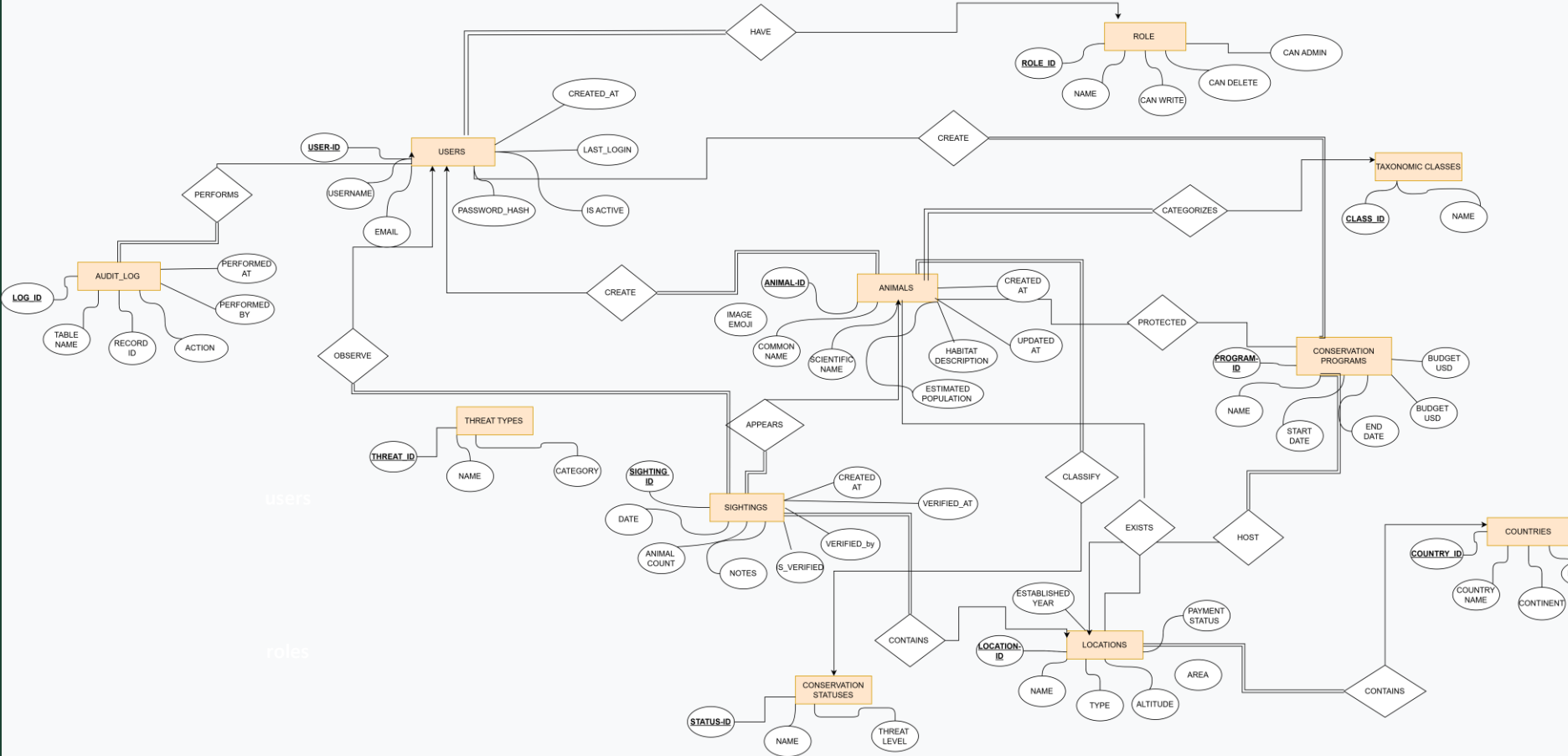
Architecture & Application Workflow



Key System Capabilities

-  Animal Profiles & Taxonomy
-  Sighting Records with Location
-  Conservation Programs
-  Role-Based Authentication
-  Audit Logs & Reporting
-  Real-Time Database Updates

Entity-Relationship Diagram



CREATE TABLE — Constraints & Keys

```
CREATE TABLE roles (
  role_id    SERIAL    PRIMARY KEY,
  role_name  VARCHAR(50) NOT NULL UNIQUE,
  can_write  BOOLEAN   NOT NULL DEFAULT FALSE,
  can_delete BOOLEAN   NOT NULL DEFAULT FALSE,
  can_admin  BOOLEAN   NOT NULL DEFAULT FALSE
);

CREATE TABLE users (
  user_id    SERIAL    PRIMARY KEY,
  username   VARCHAR(50) NOT NULL UNIQUE,
  email      VARCHAR(120) NOT NULL UNIQUE,
  password_hash VARCHAR(255) NOT NULL,
  role_id    SMALLINT   NOT NULL REFERENCES roles(role_id) ON UPDATE
CASCADE,
  is_active  BOOLEAN   NOT NULL DEFAULT TRUE,
  last_login TIMESTAMP  NULL,
  created_at TIMESTAMP  NOT NULL DEFAULT NOW()
);
```

```
CREATE TABLE conservation_statuses (
  status_id    SERIAL    PRIMARY KEY,
  iucn_code    CHAR(2)    NOT NULL UNIQUE,
  status_name  VARCHAR(50) NOT NULL,
  threat_level VARCHAR(20) NOT NULL CHECK (threat_level IN
('Low', 'Medium', 'High', 'Endangered', 'Critical'))
);
```

```
CREATE TABLE taxonomic_classes (
  class_id    SERIAL    PRIMARY KEY,
  class_name  VARCHAR(60) NOT NULL UNIQUE
);
```

```
CREATE TABLE countries (
  country_id    SERIAL    PRIMARY KEY,
  country_name  VARCHAR(80) NOT NULL UNIQUE,
  iso_code      CHAR(2)    NOT NULL UNIQUE,
  continent     VARCHAR(20) NOT NULL CHECK (continent IN
('Africa', 'Asia', 'Europe', 'North America', 'South America', 'Oceania'))
);
```

```
CREATE TABLE animals (
  animal_id    SERIAL    PRIMARY KEY,
  common_name  VARCHAR(100) NOT NULL,
  scientific_name VARCHAR(150) NOT NULL UNIQUE,
  class_id     SMALLINT   NOT NULL REFERENCES
taxonomic_classes(class_id),
  status_id    SMALLINT   NOT NULL REFERENCES
conservation_statuses(status_id),
  estimated_population INT    CHECK (estimated_population >= 0),
  habitat_description VARCHAR(300) NOT NULL DEFAULT '',
  diet_type    VARCHAR(15) NOT NULL DEFAULT 'Herbivore' CHECK
(diet_type IN ('Herbivore', 'Carnivore', 'Omnivore')),
  image_emoji  VARCHAR(10) NOT NULL DEFAULT '🐾',
  created_by   INT    NOT NULL REFERENCES users(user_id),
  created_at   TIMESTAMP  NOT NULL DEFAULT NOW(),
  updated_at   TIMESTAMP  NOT NULL DEFAULT NOW()
);
```

CREATE TABLE — Constraints & Keys

```
CREATE TABLE threat_types (
  threat_id SERIAL PRIMARY KEY,
  threat_name VARCHAR(80) NOT NULL,
  category VARCHAR(20) NOT NULL CHECK (category IN ('Habitat Loss', 'Poaching', 'Climate Change', 'Disease', 'Pollution', 'Other'));
);
```

```
CREATE TABLE animal_threats (
  animal_id INT NOT NULL REFERENCES animals(animal_id) ON DELETE CASCADE,
  threat_id SMALLINT NOT NULL REFERENCES threat_types(threat_id),
  severity VARCHAR(10) NOT NULL DEFAULT 'Moderate'
  CHECK (severity IN ('Low', 'Moderate', 'High', 'Critical')),
  PRIMARY KEY (animal_id, threat_id)
);
```

```
CREATE TABLE locations (
  location_id SERIAL PRIMARY KEY,
  location_name VARCHAR(150) NOT NULL,
  country_id SMALLINT NOT NULL REFERENCES countries(country_id),
  latitude NUMERIC(9,6) NOT NULL CHECK (latitude BETWEEN -90 AND 90),
  longitude NUMERIC(9,6) NOT NULL CHECK (longitude BETWEEN -180 AND 180),
  area_km2 NUMERIC(10,2) NOT NULL CHECK (area_km2 > 0),
  location_type VARCHAR(25) NOT NULL CHECK (location_type IN
    ('National Park', 'Game Reserve', 'Wildlife Sanctuary', 'Marine Reserve', 'UNESCO Site')),
  established_year SMALLINT NULL
);
```

```
CREATE TABLE location_species (
  location_id INT NOT NULL REFERENCES locations(location_id)
  ON DELETE CASCADE,
  animal_id INT NOT NULL REFERENCES
  animals(animal_id) ON DELETE CASCADE,
  is_primary BOOLEAN NOT NULL DEFAULT FALSE,
  resident_population INT NULL,
  PRIMARY KEY (location_id, animal_id)
);
```

```
CREATE TABLE sightings (
  sighting_id SERIAL PRIMARY KEY,
  animal_id INT NOT NULL REFERENCES animals(animal_id),
  location_id INT NULL REFERENCES locations(location_id)
  ON DELETE SET NULL,
  location_text VARCHAR(200) NOT NULL DEFAULT '',
  observer_id INT NOT NULL REFERENCES users(user_id),
  sighting_date DATE NOT NULL,
  animal_count INT NOT NULL DEFAULT 1 CHECK (animal_count >= 1),
  notes TEXT NULL,
  is_verified BOOLEAN NOT NULL DEFAULT FALSE,
  verified_by INT NULL REFERENCES users(user_id) ON DELETE
  SET NULL,
  verified_at TIMESTAMP NULL,
  created_at TIMESTAMP NOT NULL DEFAULT NOW()
);
```

CREATE TABLE — Constraints & Keys

```
CREATE TABLE conservation_programs (  
  program_id SERIAL PRIMARY KEY,  
  program_name VARCHAR(150) NOT NULL UNIQUE,  
  start_date DATE NOT NULL,  
  end_date DATE NULL,  
  budget_usd NUMERIC(14,2) CHECK (budget_usd >= 1000),  
  location_id INT NULL REFERENCES locations(location_id) ON DELETE SET NULL,  
  status VARCHAR(15) NOT NULL DEFAULT 'Planning'  
  CHECK (status IN ('Planning', 'Active', 'Completed', 'Suspended')),  
  created_by INT NOT NULL REFERENCES users(user_id)  
);  
  
CREATE TABLE program_animals (  
  program_id INT NOT NULL REFERENCES conservation_programs(program_id) ON DELETE  
  CASCADE,  
  animal_id INT NOT NULL REFERENCES animals(animal_id),  
  target_population INT NULL,  
  PRIMARY KEY (program_id, animal_id)  
);  
  
CREATE TABLE audit_log (  
  log_id BIGSERIAL PRIMARY KEY,  
  table_name VARCHAR(60) NOT NULL,  
  record_id INT NOT NULL,  
  action VARCHAR(10) NOT NULL CHECK (action IN ('INSERT', 'UPDATE', 'DELETE')),  
  performed_by INT NULL,  
  details TEXT NULL,  
  performed_at TIMESTAMP NOT NULL DEFAULT NOW()  
);
```

INSERT · UPDATE · DELETE · SELECT

INSERT

```
INSERT INTO animals (common_name, scientific_name, class_id, status_id,
estimated_population, habitat_description, diet_type, image_emoji, created_by)
VALUES
  ('African Elephant', 'Loxodonta africana', 1,3,415000,'Savannas of sub-
Saharan Africa', 'Herbivore', '🐘',1),
  ('Bengal Tiger', 'Panthera tigris tigris', 1,2, 2500,'Dense forests
of Indian subcontinent', 'Carnivore', '🐅',1),
  ('Mountain Gorilla', 'Gorilla beringei beringei', 1,1, 1063,'Virunga
volcanic mountains', 'Herbivore', '🐼',1),
  ('Giant Panda', 'Ailuropoda melanoleuca', 1,3, 1864,'Bamboo forests
of Sichuan, China', 'Herbivore', '🐼',1),
  ('Green Sea Turtle', 'Chelonia mydas', 3,2,200000,'Tropical oceans
worldwide', 'Herbivore', '🐢',2),
  ('Black Rhino', 'Diceros bicornis', 1,1, 5630,'Savannas of
eastern/southern Africa', 'Herbivore', '🐘',1),
  ('Bornean Orangutan', 'Pongo pygmaeus', 1,1,104700,'Tropical
rainforests of Borneo', 'Omnivore', '🐼',2),
  ('Snow Leopard', 'Panthera uncia', 1,3, 6500,'Mountain ranges
of Central Asia', 'Carnivore', '🐼',3),
  ('Barbary Macaque', 'Macaca sylvanus', 1,2, 8000,'Atlas Mountains
cedar forests', 'Omnivore', '🐼',3),
  ('Northern Bald Ibis', 'Geronticus eremita', 2,1, 500,'Semi-arid
cliffs, Morocco', 'Omnivore', '🐼',3);
```

UPDATE

```
UPDATE animals SET estimated_population =
430000, updated_at = NOW() WHERE common_name = 'African
Elephant';
```

```
UPDATE sightings SET animal_count = 6, notes = 'Re-
counted: mother, calf and 4 sub-adults' WHERE
sighting_id = 6;
```

```
UPDATE sightings SET is_verified = TRUE, verified_by =
2, verified_at = NOW() WHERE sighting_id = 9;
```

INSERT · UPDATE · DELETE · SELECT

SELECT

```
SELECT
    l.location_name,
    c.country_name,
    l.area_km2
FROM locations l
JOIN countries c
ON l.country_id = c.country_id;
```

```
SELECT
    s.sighting_id,
    a.common_name,
    s.sighting_date,
    s.animal_count,
    u.username AS observer
FROM sightings s
JOIN animals a
ON s.animal_id = a.animal_id
JOIN users u
ON s.observer_id = u.user_id;
```

DELETE

```
DELETE FROM location_species WHERE location_id = 8 AND
animal_id = 7;
```

Delete sightings before a certain date

```
DELETE FROM sightings
WHERE sighting_date < '2024-01-01';
```

Database Query Operations & Data Retrieval

List out info animals with conservation status

```
SELECT a.common_name, a.scientific_name, cs.status_name,  
       cs.threat_level, a.estimated_population  
FROM animals a JOIN conservation_statuses cs ON a.status_id =  
       cs.status_id  
ORDER BY cs.status_id, a.common_name;
```

common_name	scientific_name	status_name	threat_level	estimated_population
Northern Bald Ibis	Geronticus eremita	Critically Endangered	Critical	500
Barbary Macaque	Macaca sylvanus	Endangered	Endangered	8000
Bengal Tiger	Panthera tigris tigris	Endangered	Endangered	2500
Green Sea Turtle	Chelonia mydas	Endangered	Endangered	200000
African Elephant	Loxodonta africana	Vulnerable	High	430000
Giant Panda	Ailuropoda melanoleuca	Vulnerable	High	1864
Snow Leopard	Panthera uncia	Vulnerable	High	6500

List out info of animals listed more than once GROUP BY + HAVING

```
SELECT a.common_name, COUNT(s.sighting_id) AS sightings,  
       SUM(s.animal_count) AS total_seen,  
       MAX(s.sighting_date) AS last_seen  
FROM animals a LEFT JOIN sightings s ON a.animal_id =  
       s.animal_id  
GROUP BY a.animal_id, a.common_name  
HAVING COUNT(s.sighting_id) > 1  
ORDER BY sightings DESC;
```

common_name	sightings	total_seen	last_seen
African Elephant	2	73	2025-03-18

Database Query Operations & Data Retrieval

Full sighting details alongside observer role using INNER JOIN

```
SELECT s.sighting_id, a.common_name,  
       COALESCE(l.location_name, s.location_text) AS location,  
       u.username AS observer, r.role_name, s.sighting_date,  
       s.animal_count, s.is_verified  
FROM sightings s  
INNER JOIN animals a ON s.animal_id = a.animal_id  
INNER JOIN users u ON s.observer_id = u.user_id  
INNER JOIN roles r ON u.role_id = r.role_id  
LEFT JOIN locations l ON s.location_id = l.location_id  
ORDER BY s.sighting_date DESC;
```

sighting_id	common_name	location	observer	role_name	sighting_date	animal_count	is_verified
7	Barbary Macaque	Toubkal National Park	Adam	Administrator	2025-03-20	15	false
4	African Elephant	Masai Mara National Reserve	Majdouline	Researcher	2025-03-18	28	true
8	Northern Bald Ibis	Souss-Massa National Park	Majdouline	Researcher	2025-03-17	8	true
1	African Elephant	Amboseli National Park	Adam	Administrator	2025-03-15	45	true
5	Giant Panda	Sichuan Panda Sanctuary	Hanae	Researcher	2025-03-12	2	true
2	Bengal Tiger	Ranthambore National Park	Bachair	Field Ranger	2025-03-10	3	true
6	Black Rhino	Etosha National Park	Farid	Field Ranger	2025-03-08	6	true
3	Mountain Gorilla	Volcanoes National Park	Omar	Administrator	2025-03-05	12	true
10	Bornean Orangutan	Kinabalu Park, Borneo	Hanae	Researcher	2025-02-25	6	true
9	Snow Leopard	Hindu Kush Mountains	Bachair	Field Ranger	2025-02-20	2	true

List out animals with their program coverage using FULL OUTER JOIN

```
SELECT a.common_name, cp.program_name  
FROM animals a  
FULL OUTER JOIN program_animals pa ON a.animal_id =  
pa.animal_id  
FULL OUTER JOIN conservation_programs cp ON  
pa.program_id = cp.program_id  
ORDER BY a.common_name;
```

common_name	program_name
African Elephant	Elephant Corridor Kenya
Barbary Macaque	Atlas Primate Shield
Bengal Tiger	Project Bengal
Black Rhino	NULL
Bornean Orangutan	NULL
Giant Panda	NULL
Green Sea Turtle	NULL

Database Query Operations & Data Retrieval

List out animals info with an active program

```
SELECT common_name, scientific_name FROM animals
WHERE animal_id IN (
  SELECT pa.animal_id FROM program_animals pa
  JOIN conservation_programs cp ON pa.program_id = cp.program_id
  WHERE cp.status = 'Active');
```

common_name	scientific_name
Northern Bald Ibis	Geronticus eremita
Barbary Macaque	Macaca sylvanus
African Elephant	Loxodonta africana
Bengal Tiger	Panthera tigris tigris

Views • Triggers • Functions • Indexes

VIEW

- virtual dashboard view that combines wildlife sightings, animal details, locations, countries, observers, and verification information into one unified

```
CREATE OR REPLACE VIEW v_sighting_dashboard AS
SELECT s.sighting_id, a.common_name AS animal, a.image_emoji, cs.threat_level,
       COALESCE(l.location_name, s.location_text) AS location, co.country_name,
       s.sighting_date, s.animal_count, u.username AS observer,
       s.is_verified, v.username AS verified_by_user
FROM sightings s
JOIN animals a          ON s.animal_id = a.animal_id
JOIN conservation_statuses cs ON a.status_id = cs.status_id
JOIN users u            ON s.observer_id = u.user_id
LEFT JOIN locations l    ON s.location_id = l.location_id
LEFT JOIN countries co   ON l.country_id = co.country_id
LEFT JOIN users v        ON s.verified_by = v.user_id;
```

- view that displays each animal's classification, conservation status, population, associated threats, and total threat count in a single summarized profile.

```
CREATE OR REPLACE VIEW v_animal_threat_profile AS
SELECT a.animal_id, a.common_name, cs.status_name, cs.threat_level,
       a.estimated_population,
       STRING_AGG(tt.threat_name || ' [' || at2.severity || ']', ';' )
         ORDER BY at2.severity DESC) AS threats,
       COUNT(tt.threat_id) AS threat_count
FROM animals a
JOIN taxonomic_classes tc    ON a.class_id = tc.class_id
JOIN conservation_statuses cs ON a.status_id = cs.status_id
LEFT JOIN animal_threats at2  ON at2.animal_id = a.animal_id
LEFT JOIN threat_types tt     ON tt.threat_id = at2.threat_id
GROUP BY a.animal_id, a.common_name, tc.class_name, cs.status_name,
cs.threat_level, a.estimated_population;
```

TRANSACTION

- T Grants logging in and verifying a sighting automatically

```
BEGIN;

SAVEPOINT sp_sighting;

INSERT INTO sightings (animal_id, location_id,
                      location_text, observer_id, sighting_date, animal_count,
                      notes)
VALUES (2, 5, 'Ranthambore Zone 4', 5, CURRENT_DATE, 1,
'Solo male at waterhole');

UPDATE sightings SET is_verified=TRUE, verified_by=2,
verified_at=NOW()
WHERE sighting_id = (SELECT MAX(sighting_id) FROM
sightings);

UPDATE location_species SET resident_population =
resident_population + 1
WHERE location_id = 5 AND animal_id = 2;

COMMIT;
```

Views • Triggers • Functions • Indexes

INDEX

- These indexes improve database performance by speeding up searches, filtering, joins, and full-text search operations in the wildlife conservation system.

```
CREATE INDEX idx_sightings_animal
  ON sightings(animal_id);
CREATE INDEX idx_animals_fts
  ON animals USING GIN(to_tsvector
    ('english', name || ' ' || species)) CREATE INDEX
idx_sightings_date      ON sightings (sighting_date);
CREATE INDEX idx_sightings_animal  ON sightings (animal_id,
sighting_date);
CREATE INDEX idx_sightings_observer ON sightings (observer_id);
CREATE INDEX idx_animals_status    ON animals (status_id);
CREATE INDEX idx_animals_name      ON animals (common_name);
CREATE INDEX idx_locations_country ON locations (country_id);
CREATE INDEX idx_users_email       ON users (email);
CREATE INDEX idx_audit_table       ON audit_log (table_name,
performed_at);
CREATE INDEX ft_sightings_notes    ON sightings USING GIN
(to_tsvector('english', COALESCE(notes,'')));
CREATE INDEX ft_animals_habitat    ON animals USING GIN
(to_tsvector('english', habitat_description));
```

TRIGGER

- A trigger function that auto-escalates IUCN status when population drops

```
CREATE OR REPLACE FUNCTION trg_population_escalate()
RETURNS TRIGGER LANGUAGE plpgsql AS $$
DECLARE v_new_status INT;
BEGIN
  IF NEW.estimated_population IS DISTINCT FROM
  OLD.estimated_population THEN
    v_new_status := CASE
      WHEN NEW.estimated_population < 250 THEN 1 -- CR
      WHEN NEW.estimated_population < 2500 THEN 2 -- EN
      WHEN NEW.estimated_population < 10000 THEN 3 -- VU
      ELSE OLD.status_id
    END;
    IF v_new_status <> OLD.status_id THEN
      NEW.status_id := v_new_status;
      INSERT INTO wildlife.audit_log (table_name, record_id,
        action, details)
        VALUES ('animals', NEW.animal_id, 'UPDATE',
          'Status escalated: pop ' ||
            OLD.estimated_population || '->' || NEW.estimated_population);
    END IF;
  END IF;
  RETURN NEW;
END;
$$;
```

PL/pgSQL — Stored Procedures & Functions

```
CREATE OR REPLACE FUNCTION sp_log_sighting(
    p_animal_id INT, p_location TEXT,
    p_observer INT, p_date DATE,
    p_count INT, p_notes TEXT
) RETURNS TEXT LANGUAGE plpgsql AS $$
DECLARE
    v_loc_id INT;
    v_new_id INT;
BEGIN
    IF NOT EXISTS (SELECT 1 FROM wildlife.animals WHERE animal_id = p_animal_id) THEN
        RETURN 'ERROR: Animal not found.';
    END IF;
    IF p_date > CURRENT_DATE THEN
        RETURN 'ERROR: Future date not allowed.';
    END IF;

    SELECT location_id INTO v_loc_id FROM wildlife.locations
    WHERE location_name ILIKE '%' || p_location || '%' LIMIT 1;

    INSERT INTO wildlife.sightings (animal_id, location_id, location_text, observer_id,
    sighting_date, animal_count, notes)
    VALUES (p_animal_id, v_loc_id, p_location, p_observer, p_date, p_count, p_notes)
    RETURNING sighting_id INTO v_new_id;

    INSERT INTO wildlife.audit_log (table_name, record_id, action, performed_by, details)
    VALUES ('sightings', v_new_id, 'INSERT', p_observer,
    'Animal #' || p_animal_id || ' count=' || p_count);

    RETURN 'SUCCESS: Sighting #' || v_new_id || ' logged.';
EXCEPTION WHEN OTHERS THEN
    RETURN 'ERROR: ' || SQLERRM;
END;
$$;
```

A function logging sightings with validation

```
CREATE OR REPLACE FUNCTION fn_coords(p_lat NUMERIC, p_lng
NUMERIC)
RETURNS TEXT LANGUAGE sql IMMUTABLE AS $$
    SELECT ABS(ROUND(p_lat,4)) || '°' || CASE WHEN p_lat >=
0 THEN 'N' ELSE 'S' END
    || ' ' ||
    ABS(ROUND(p_lng,4)) || '°' || CASE WHEN p_lng >=
0 THEN 'E' ELSE 'W' END;
$$$;
```

A function rereturning human-readable coordinates

PL/pgSQL — Stored Procedures & Functions

```
CREATE OR REPLACE FUNCTION fn_country_report(p_country TEXT)
RETURNS TABLE (
  country      TEXT, parks BIGINT, total_km2 NUMERIC,
  species      BIGINT, sightings BIGINT, critical_species BIGINT
) LANGUAGE sql STABLE AS $$
  SELECT c.country_name,
         COUNT(DISTINCT l.location_id),
         ROUND(SUM(l.area_km2)),
         COUNT(DISTINCT ls.animal_id),
         COUNT(DISTINCT s.sighting_id),
         COUNT(DISTINCT CASE WHEN cs.threat_level = 'Critical' THEN a.animal_id END)
  FROM wildlife.countries c
  JOIN wildlife.locations l      ON l.country_id  = c.country_id
  LEFT JOIN wildlife.location_species ls ON ls.location_id = l.location_id
  LEFT JOIN wildlife.animals a      ON a.animal_id  = ls.animal_id
  LEFT JOIN wildlife.conservation_statuses cs ON cs.status_id = a.status_id
  LEFT JOIN wildlife.sightings s      ON s.location_id = l.location_id
  WHERE c.country_name = p_country
  GROUP BY c.country_name;
$$;
```

-- Country wildlife state as a set-returning function

```
CREATE OR REPLACE PROCEDURE DeleteAnimalSecure(
  p_animal_id INT,
  p_user_id INT
)
LANGUAGE plpgsql
AS $$
DECLARE
  v_can_delete BOOLEAN;
BEGIN
  SELECT r.can_delete
  INTO v_can_delete
  FROM users u
  JOIN roles r ON u.role_id = r.role_id
  WHERE u.user_id = p_user_id;

  IF NOT v_can_delete THEN
    RAISE EXCEPTION 'Permission denied';
  END IF;

  DELETE FROM animals
  WHERE animal_id = p_animal_id;

  INSERT INTO audit_log(
    table_name,
    record_id,
    action,
    performed_by,
    details
  )
  VALUES(
    'animals',
    p_animal_id,
    'DELETE',
    p_user_id,
    'Animal deleted securely'
  );

  RAISE NOTICE 'Animal deleted';
END;
$$;
```

A procedure checking whether the user can delete an animal or not

Cloud Database Connection & Auth



```
// FILE: supabase.js → Database Connection
const supabaseUrl='https://izttjdcxsdmktxdnhl1n.supabase.co'; const supabaseKey='YOUR_SUPABASE_ANON_KEY';
const supabaseClient=supabase.createClient(supabaseUrl,supabaseKey);

// FILE: script.js → Authentication System
async function authenticateUser(email,password){ const {data,error}=await
supabaseClient.from('users').select('user_id,username,email,password_hash,role_id,roles(role_name)').eq('
email',email).eq('password_hash',password).single(); if(error||!data){ return
{success:false,error:'Invalid email or password'}}; } return
{success:true,user:{id:data.user_id,name:data.username,email:data.email,role:data.roles.role_name}}; }

// FILE: animals.js → Fetch Animals
async function fetchAnimals(){ const {data,error}=await
supabaseClient.from('animals').select('animal_id,common_name,scientific_name,estimated_population,habitat
_description,image_emoji'); if(error){ console.error(error); return []; } return data; }

// FILE: animals.js → Insert Animal
await
supabaseClient.from('animals').insert([{common_name:commonName,scientific_name:scientificName,class_id:1,
status_id:1,estimated_population:population,habitat_description:habitat,diet_type:'Carnivore',image_emoji
:imageEmoji,created_by:1}]);

// FILE: animals.js → Delete Animal
await supabaseClient.from('animals').delete().eq('animal_id',id);

// FILE: index.html → Load Scripts
<script src="https://cdn.jsdelivr.net/npm/@supabase/supabase-js@2"></script> <script
src="supabase.js"></script> <script src="script.js"></script>
```



Cloud Hosted

Zero-infrastructure
PostgreSQL database on
Supabase cloud



Row Level Security

Policies ensure
users only see their
authorized data



JWT Auth

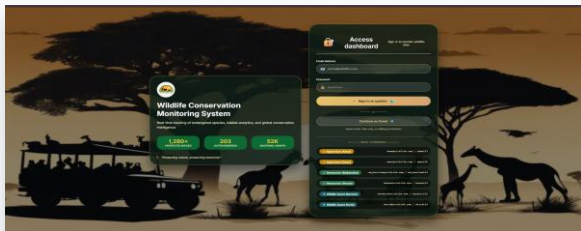
Supabase Auth issues JWT
tokens for each session



Real-time API

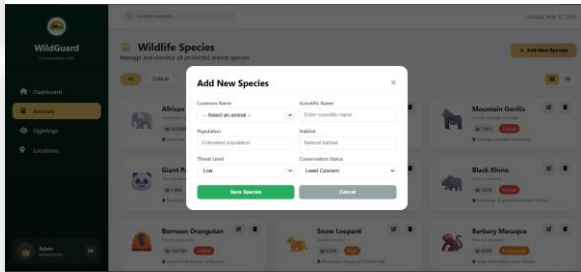
WebSocket subscriptions
for live data updates

Frontend Screens & User Experience



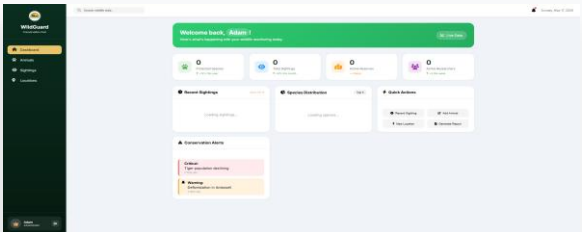
Login Page

Supabase Auth with email/password. JWT stored in session. Redirects by role after login.



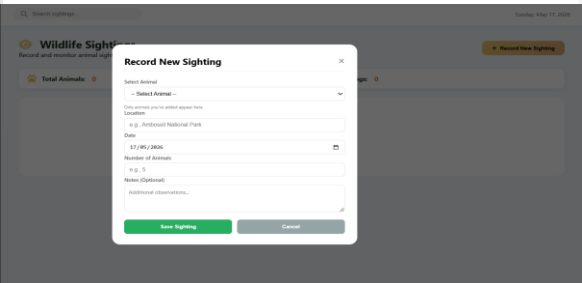
Add Animal Modal

Form with validation — species, status dropdown (CHECK-constrained), location picker.



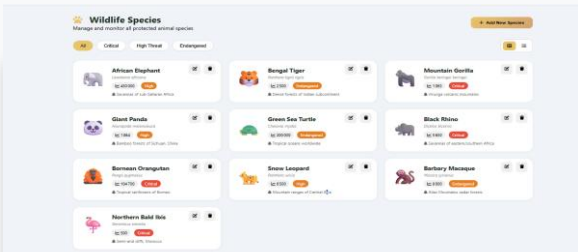
Dashboard

Real-time KPIs: total animals, sightings today, active programs, endangered count.



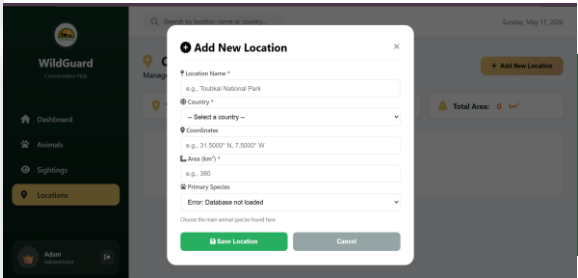
Sightings Map

Grid of recent sightings with animal name, location, date, and reporter info.



Animals List

Paginated table with search, filter by status/species, and inline status badges.



Role-Based Access

Admins: full CRUD. Rangers: add sightings. Researchers: read-only access to reports.

TESTING THE UI WITH THE DATABASE SIDE

Real-Time Species Insertion into PostgreSQL Database

[SCREENSHOT]

Cette page indique
✓ Animal saved successfully!

OK

Add New Species

Common Name: Cheetah Scientific Name: Acinonyx jubatus

Population: 7100 Habitat: African grasslands

Threat Level: High Conservation Status: Least Concern

Save Species Cancel

FRONTEND APPLICATION

The user enters animal information through the Animals Management interface.

[SCREENSHOT]

```
SQL SELECT * FROM animals;
```

animal_id	common_name	scientific_name	class_id	status_id	estimated_pop	habitat_description
12	Cheetah	Acinonyx jubatus	1	3	7100	African grasslands

SUPABASE SIDE

The new animal record is stored permanently in the cloud PostgreSQL database.
The inserted record can be viewed directly in the Supabase Table Editor or SQL queries.

Cheetah
Acinonyx jubatus

7100 High

African grasslands

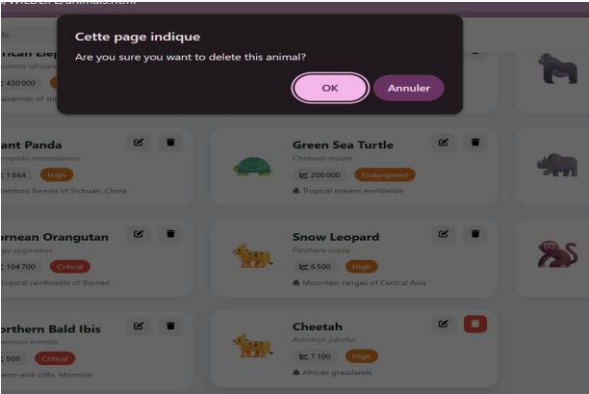
THE ANIMAL PAGE

The application automatically refreshes and displays the new animal in the interface.

TESTING THE UI WITH THE DATABASE SIDE

Secure Species Deletion Using Supabase & PostgreSQL

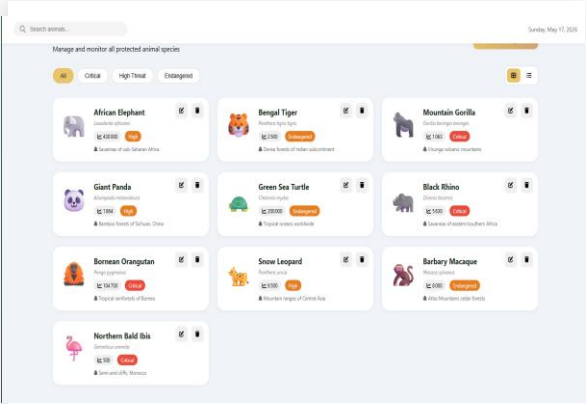
[SCREENSHOT]



Species Selection

The administrator selects a species to remove from the Animals interface.

[SCREENSHOT]



Automatic Interface Refresh

The application updates the interface instantly to reflect the deletion.

	animal_id	common_name	scientific_name	class_id	status_id	estimated_pop	habitat_description
5		Green Sea Turtle	Chelonia mydas	3	2	200000	Tropical oceans worldwide
6		Black Rhino	Diceros bicornis	1	1	5038	Savannas of eastern/southern Africa
7		Bornean Orangutan	Pongo pygmaeus	1	1	104700	Tropical rainforests of Borneo
8		Snow Leopard	Panthera uncia	1	3	6500	Mountain ranges of Central Asia
9		Barbary Macaque	Macaca sylvanus	1	2	8000	Atlas Mountains cedar forests
10		Northern Bald Ibis	Geronticus eremita	2	1	500	Semi-arid cliffs, Morocco
11		African Elephant	Loxodonta africana	1	3	430000	Savannas of sub-Saharan Africa

Cloud Database Synchronization

The deletion is immediately visible in the Supabase Table Editor and SQL queries.

Tools & Workflow



VS Code

Primary IDE

IntelliSense, Prettier formatting, ESLint linting, and the Supabase extension for schema browsing.



Live Server

Local Dev Server

Hot-reload browser preview during frontend development — zero build step required.



Supabase Dashboard

DB Management

Visual table editor, SQL editor, auth management, and real-time log streaming.



SQL Editor

Query Testing

Write and test complex queries, run migrations, inspect query plans with EXPLAIN ANALYZE.



GitHub

Version Control

Feature-branch workflow. Pull requests reviewed before merging. README with setup guide.



DRAW.IO

Advanced DB Tools

Used for ER diagram visualization, import/export, and backup operations.

System Achievements & Verified Outputs

+10

Database Tables

Fully normalized 3NF

12+

SQL Objects

Views, triggers, functions

4

User Roles

Admin · Guard · Researcher · Guest

100%

CRUD Coverage

All entities fully operable

Project Summary & Key Achievements

- ✓ Designed and deployed a fully normalized PostgreSQL database with 8 entities in Supabase cloud.
- ✓ Implemented advanced SQL features: 3 views, 5 PL/pgSQL blocks, 4 triggers, 6 indexes, and full-text search.
- ✓ Built role-based authentication (Admin/Ranger/Researcher) with Row Level Security policies.
- ✓ Developed a responsive frontend in HTML/CSS/JavaScript with real-time Supabase integration.
- ✓ Applied 3NF normalization, referential integrity, and proper constraint design throughout.
- ✓ Successfully completed full CRUD operations with audit logging, automated triggers, and cascade rules.

"This project transformed our understanding of database systems from theory into real-world practice — integrating cloud infrastructure, security, and full-stack development into a single cohesive system."



Thank You

Wildlife Conservation Management System

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Cyprus International University | CMPE344 · 2026

GITHUB LINK

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